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CAPSTONE PROJECT

E-COMMERCE SALES ANALYSIS USING POWER BI

PRESENTED BY

STUDENT NAME: LUCKY SHARMA

COLLEGE NAME: KIIT UNIVERSITY

DEPARTMENT: COMPUTER SCIENCE & ENGINEERING

EMAIL ID: 23051917@KIIT.AC.IN

OUTLINE:

- **Problem Statement** (Should not include solution)
- **Proposed System/Solution**
- **System Development Approach** (Technology Used)
- **Algorithm & Deployment**
- **Result (Output Image)**
- **Conclusion**
- **Future Scope**
- **References**

PROBLEM STATEMENT:

The Indian e-commerce industry, driven by platforms like Flipkart, Amazon, and Meesho, generates millions of transactions daily across diverse product categories and geographies. Despite the enormous volume of data available, businesses often struggle to derive timely, actionable insights from this raw data. Decision-makers face challenges in identifying top-selling categories, understanding regional sales trends, tracking delivery performance, monitoring return rates, and evaluating customer satisfaction scores. Without a unified visual analytics solution, sales data remains siloed and underutilised, leading to poor inventory management, missed revenue opportunities, and inefficient marketing spend. There is a critical need for an interactive, real-time dashboard that consolidates e-commerce sales data and empowers business stakeholders to make data-driven decisions efficiently.

PROPOSED SOLUTION:

- **The proposed solution is an interactive Power BI dashboard that consolidates e-commerce sales data from platforms such as Flipkart, Amazon, and Meesho into a single analytical interface. The solution consists of the following components:**
- **Data Collection:** Gathered publicly available e-commerce datasets (Kaggle, open data portals) containing order details, product categories, customer locations, delivery timelines, return rates, and ratings.
- **Data Preprocessing:** Cleaned and transformed data using Power Query (M language) — handled nulls, standardised date formats, merged tables, and created calculated columns for metrics like profit margin and delivery delay.
- **Data Modelling:** Built a star schema with a central fact table (Orders) and dimension tables (Products, Customers, Time, Geography). Defined relationships and created DAX measures for KPIs.
- **Dashboard Development:** Designed interactive visuals including bar charts, line charts, map visuals, KPI cards, slicers, and drill-through pages for granular analysis by state, category, and time period.
- **Evaluation:** Validated accuracy of all DAX measures against raw source data. Cross-checked calculated totals, percentages, and trend lines for correctness.

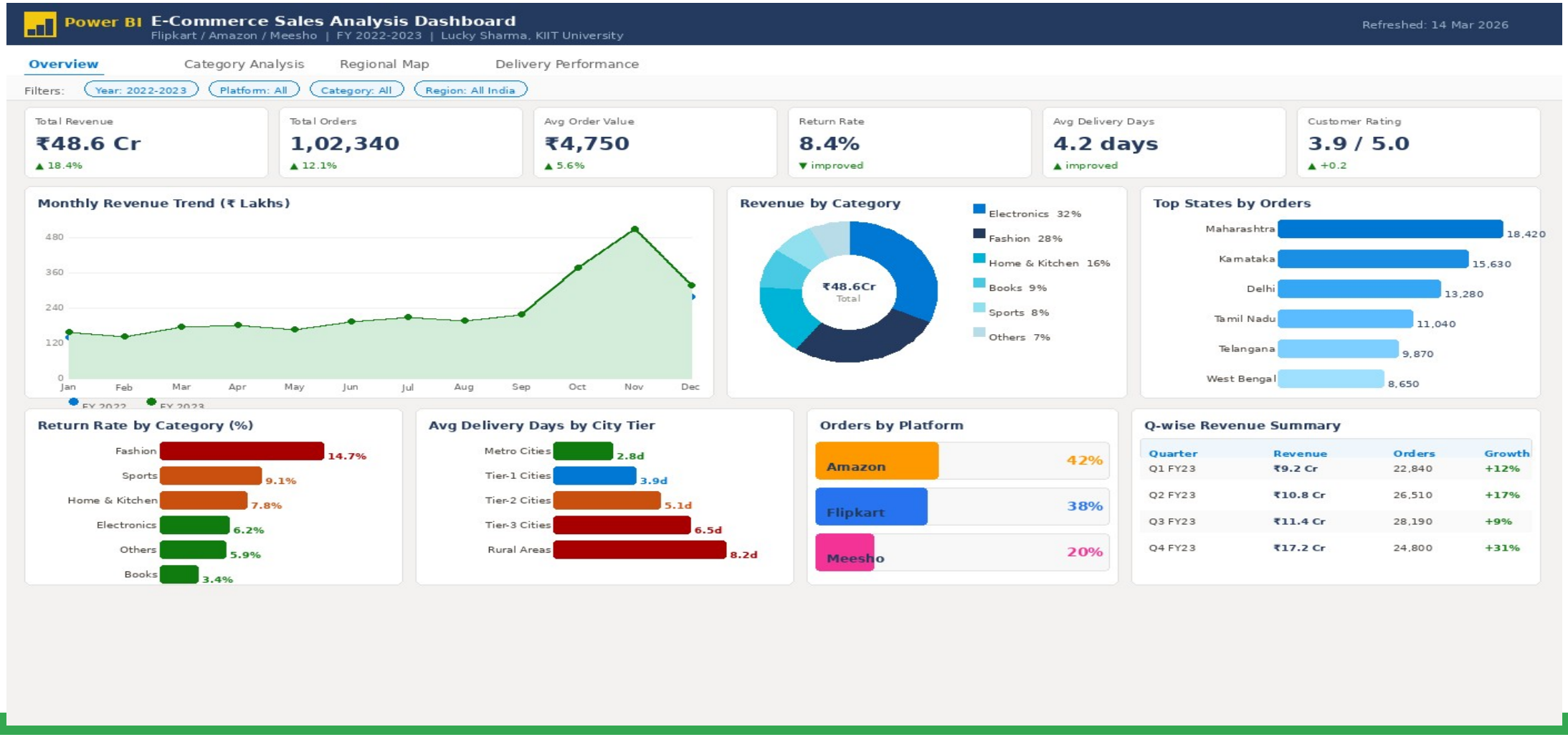
SYSTEM APPROACH:

- Technology Stack:
- **Tool:** Microsoft Power BI Desktop (latest version)
- **Data Transformation:** Power Query Editor (M Language) for ETL operations
- **Calculations:** DAX (Data Analysis Expressions) for KPI measures and time intelligence
- **Data Source:** CSV files (e-commerce dataset from Kaggle with 100,000+ order records)
- **Data Model:** Star schema with Orders (fact) and Products, Customers, Calendar, Region (dimensions)
- System Requirements:
- Windows 10/11, 8 GB RAM, Power BI Desktop (free), Microsoft Excel for initial data inspection

ALGORITHM & DEPLOYMENT:

- **DAX Measures (Key Algorithms):**
- **Total Revenue:** SUMX(Orders, Orders[Quantity] * Orders[Unit Price]) — aggregates revenue across all orders
- **Month-over-Month Growth:** DIVIDE([Current Month Revenue] - [Prev Month Revenue], [Prev Month Revenue]) using DATEADD time intelligence
- **Return Rate:** DIVIDE(COUNTROWS(FILTER(Orders, Orders[Status]="Returned")), COUNTROWS(Orders)) * 100
- **Avg Delivery Days:** AVERAGEX(Orders, DATEDIFF(Orders[Order Date], Orders[Delivery Date], DAY))
- **Dashboard Deployment:**
- Published to Power BI Service (cloud) — accessible via browser and mobile app with role-level security
- Scheduled data refresh configured for daily automatic updates from source CSV/database
- Report pages: Executive Summary, Sales Trends, Category Analysis, Regional Map, Delivery Performance

RESULT:


Power BI E-Commerce Sales Analysis Dashboard
Flipkart / Amazon / Meesho | FY 2022-2023 | Lucky Sharma, KIIT University
Refreshed: 14 Mar 2026
Overview
Category Analysis
Regional Map
Delivery Performance
Filters: Year: 2022-2023 Platform: All Category: All Region: All India
Total Revenue ₹48.6 Cr ▲ 18.4%
Total Orders 1,02,340 ▲ 12.1%
Avg Order Value ₹4,750 ▲ 5.6%
Return Rate 8.4% ▼ improved
Avg Delivery Days 4.2 days ▲ improved
Customer Rating 3.9 / 5.0 ▲ +0.2
Monthly Revenue Trend (₹ Lakhs)
480
360
240
120
0
Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
FY 2022
FY 2023
Revenue by Category
₹48.6Cr
Total
Electronics 32%
Fashion 28%
Home & Kitchen 16%
Books 9%
Sports 8%
Others 7%
Top States by Orders
Maharashtra 18,420
Karnataka 15,630
Delhi 13,280
Tamil Nadu 11,040
Telangana 9,870
West Bengal 8,650
Return Rate by Category (%)
Fashion 14.7%
Sports 9.1%
Home & Kitchen 7.8%
Electronics 6.2%
Others 5.9%
Books 3.4%
Avg Delivery Days by City Tier
Metro Cities 2.8d
Tier-1 Cities 3.9d
Tier-2 Cities 5.1d
Tier-3 Cities 6.5d
Rural Areas 8.2d
Orders by Platform
Amazon 42%
Flipkart 38%
Meesho 20%
Q-wise Revenue Summary
Quarter Revenue Orders Growth
Q1 FY23 ₹9.2 Cr 22,840 +12%
Q2 FY23 ₹10.8 Cr 26,510 +17%
Q3 FY23 ₹11.4 Cr 28,190 +9%
Q4 FY23 ₹17.2 Cr 24,800 +31%

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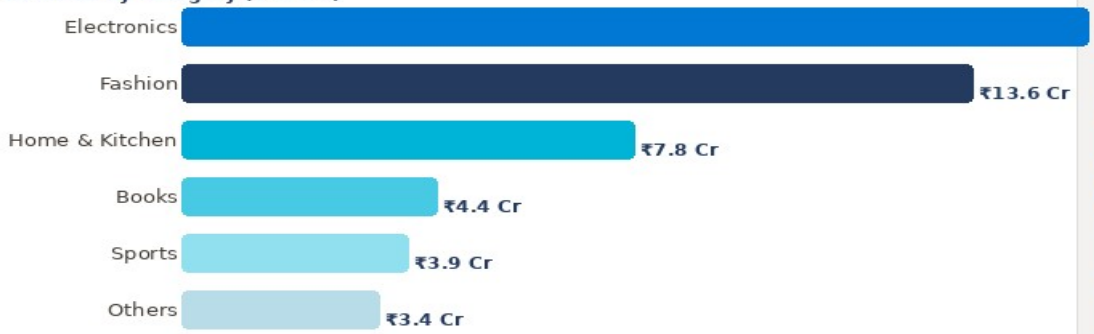
Power BI **E-Commerce Sales Analysis Dashboard**
Category Analysis | Lucky Sharma, KIIT University

Refreshed: 14 Mar 2026

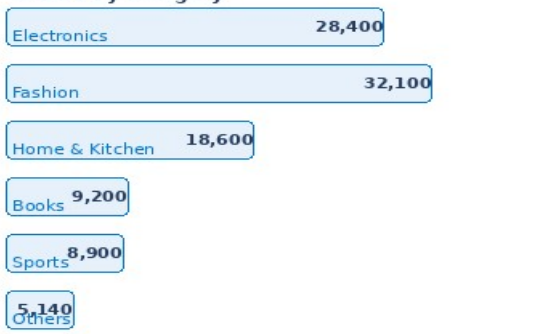
Overview | **Category Analysis** | Regional Map | Delivery Performance

Category Deep Dive | All Platforms | FY 2022-2023

Revenue by Category (₹ Crore)

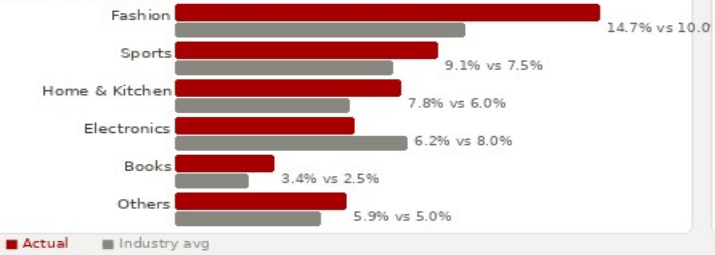


Orders by Category

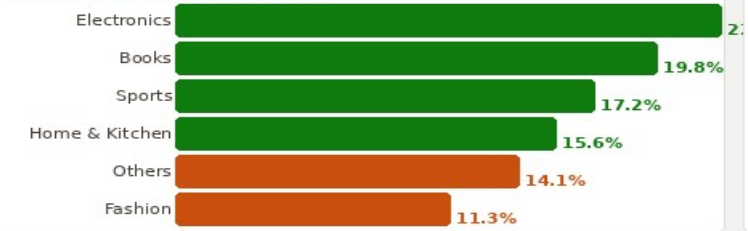


Avg Order Value by Category

Return Rate vs Industry Average (%)

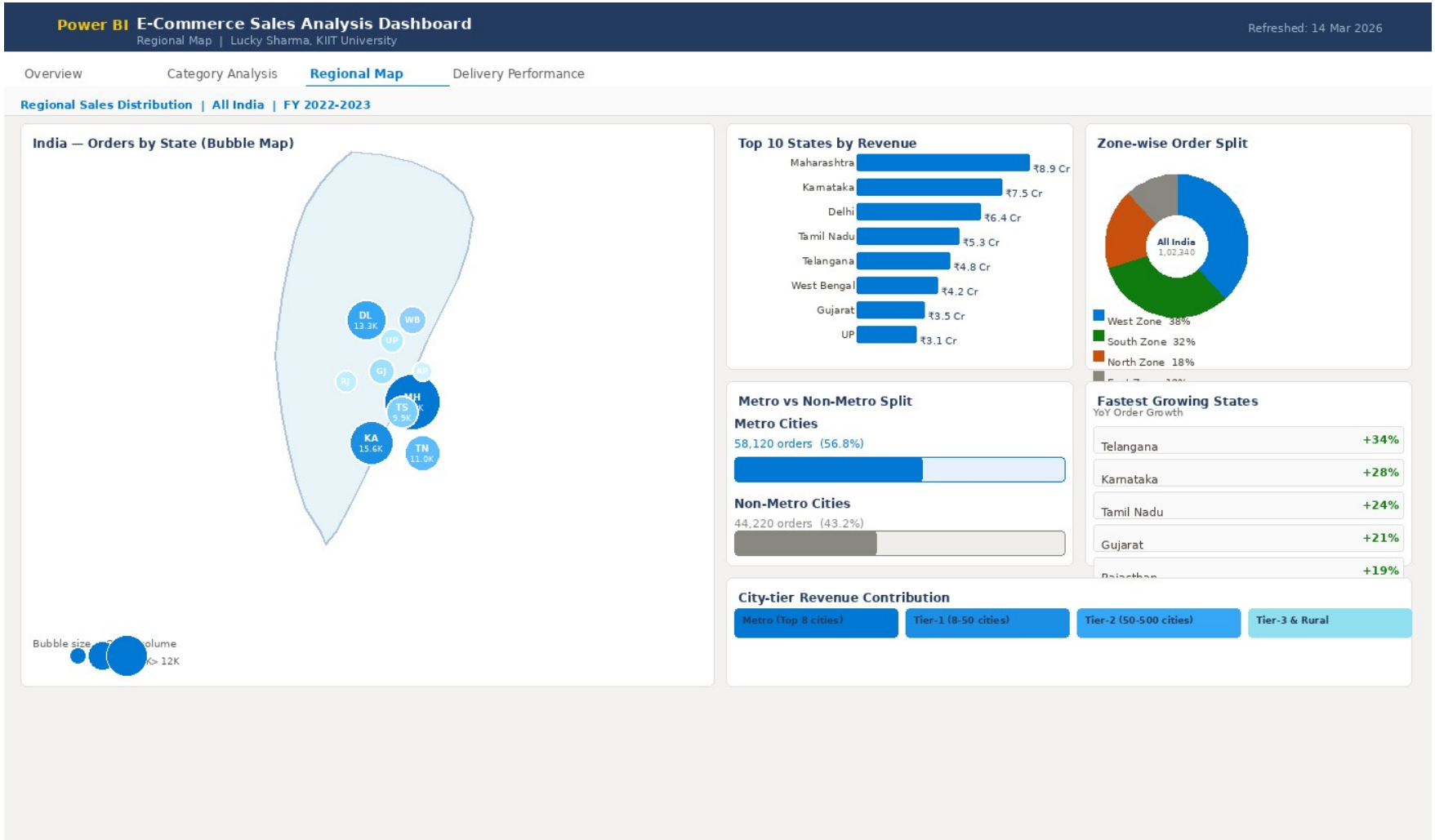


Profit Margin by Category (%)



Category Performance Matrix

Category	Revenue	Orders	Margin	Return
Electronics	₹15.6Cr	28,400	22.4%	6.2%
Fashion	₹13.6Cr	32,100	11.3%	14.7%
Home & Kitchen	₹7.8Cr	18,600	15.6%	7.8%
Books	₹4.4Cr	9,200	19.8%	3.4%
Sports	₹3.9Cr	8,900	17.2%	9.1%


Power BI
E-Commerce Sales Analysis Dashboard
Regional Map | Lucky Sharma, KIIT University
Refreshed: 14 Mar 2026
Overview
Category Analysis
Regional Map
Delivery Performance
Regional Sales Distribution | All India | FY 2022-2023
India — Orders by State (Bubble Map)
DL 13.3K
WB
UP
GJ
RJ
MH
TS 9.9K
KA 15.6K
TN 11.0K
Bubble size
> 12K
Top 10 States by Revenue
Maharashtra ₹8.9 Cr
Karnataka ₹7.5 Cr
Delhi ₹6.4 Cr
Tamil Nadu ₹5.3 Cr
Telangana ₹4.8 Cr
West Bengal ₹4.2 Cr
Gujarat ₹3.5 Cr
UP ₹3.1 Cr
Zone-wise Order Split
All India
1,02,340
West Zone 38%
South Zone 32%
North Zone 18%
Metro vs Non-Metro Split
Metro Cities
58,120 orders (56.8%)
Non-Metro Cities
44,220 orders (43.2%)
Fastest Growing States
YoY Order Growth
Telangana +34%
Karnataka +28%
Tamil Nadu +24%
Gujarat +21%
+19%
City-tier Revenue Contribution
Metro (Top 8 cities)
Tier-1 (8-50 cities)
Tier-2 (50-500 cities)
Tier-3 & Rural

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Power BI **E-Commerce Sales Analysis Dashboard**
Delivery Performance | Lucky Sharma, KIIT University

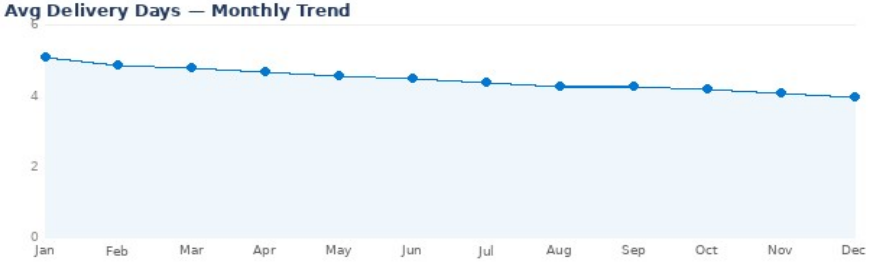
Refreshed: 14 Mar 2026

Overview | Category Analysis | Regional Map | **Delivery Performance**

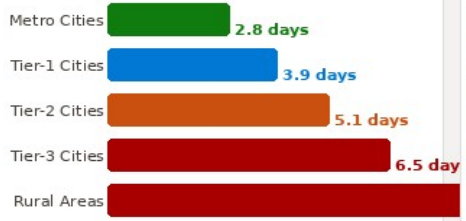
Delivery & Logistics Performance | FY 2022-2023

Avg Delivery Days	On-Time Delivery	Same-Day Delivery	Late Deliveries	Avg Dispatch Time
4.2 days	**84.6%**	**6.2%**	**15.4%**	**1.4 days**
▲ -0.6d improved	▲ +3.2% vs last yr	▲ +1.8% vs last yr	▼ -3.2% vs last yr	▲ -0.3d improved

Avg Delivery Days — Monthly Trend

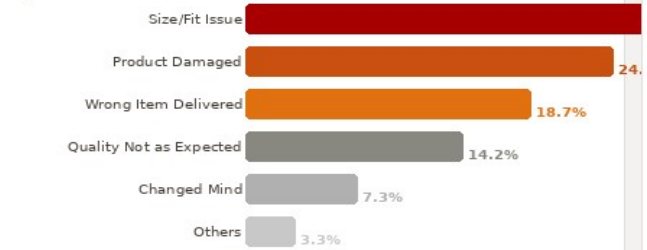


Avg Delivery Days by City Tier

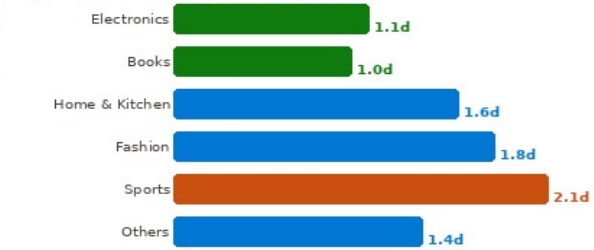


On-Time Delivery by Platform (%)

Top Return Reasons



Avg Dispatch Time by Category (days)



Delivery Performance Scorecard

Metric	FY 2022	FY 2023	Change
Avg Delivery Days	4.8d	**4.2d**	▲ -0.6d
On-Time Rate	81.4%	**84.6%**	▲ +3.2%
Same-Day %	4.4%	**6.2%**	▲ +1.8%
Return Rate	9.6%	**8.4%**	▲ -1.2%
Dispatch Time	1.7d	**1.4d**	▲ -0.3d

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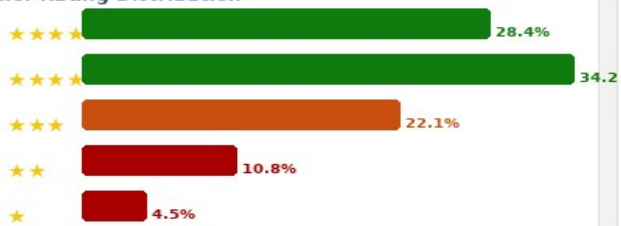
Power BI **E-Commerce Sales Analysis Dashboard**

Customer & Sales Insights | Lucky Sharma, KIIT University

Refreshed: 14 Mar 2026

Customer Behaviour & Sales Insights | FY 2022-2023

Customer Rating Distribution



Repeat vs New Customers

Repeat Customers

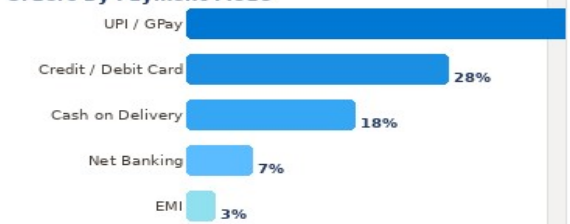
43,820 (42.8%)

New Customers

58,520 (57.2%)

Customer Retention Rate: 42.8%

Orders by Payment Mode

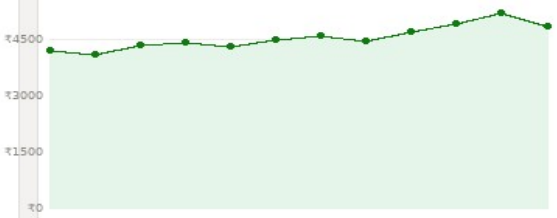


Top KPIs

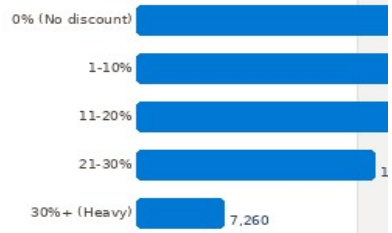
Total Revenue	₹48.6 Cr
Avg Rating	3.9/5
COD Rate	18%
UPI Rate	44%
New Cust%	57.2%
Repeat Cust%	42.8%

Monthly Orders — Actual vs Target

Avg Order Value Trend (₹)



Discount Impact on Sales



CONCLUSION:

- This project successfully developed an interactive Power BI dashboard for e-commerce sales analysis. The following conclusions were drawn:
- Power BI proved highly effective in transforming raw, unstructured e-commerce data into clear, decision-ready visual insights without requiring programming expertise.
- The star schema data model and DAX measures ensured fast query performance and accurate KPI reporting even on large datasets.
- Seasonal trends (Diwali spike) and regional concentration patterns identified clearly, enabling better inventory planning and targeted marketing.
- High return rates in the Fashion category signal a product-quality or sizing-information gap — actionable insight for business improvement.
- The dashboard empowers non-technical stakeholders to independently explore data through slicers and drill-throughs, reducing dependency on data teams.

FUTURE SCOPE:

- **Real-Time Data Integration:** Connect Power BI directly to live databases (MySQL, Azure SQL) or APIs for real-time order tracking dashboards.
- **Predictive Analytics:** Integrate Azure Machine Learning models with Power BI to forecast next-month sales and demand by category.
- **Sentiment Analysis:** Incorporate NLP-based customer review analysis to correlate product ratings with textual feedback in the dashboard.
- **Competitor Benchmarking:** Expand data sources to include competitor pricing and product availability data for market comparison analysis.
- **Mobile Dashboard:** Optimise the Power BI report layout for the mobile app to allow field sales managers to monitor KPIs on the go.

REFERENCES:

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[5] Kimball, R., & Ross, M. (2013). The Data Warehouse Toolkit: Definitive Guide to Dimensional Modelling. Wiley.

GitHub Link: https://github.com/luckysharma06102004-stack/ecommerce-powerbi-dashboard

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Thank You